Lecture 12: Cross-Sectional Analysis – Aggregation

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The aggregation problem

- Last time: **micro experiments** give inputs to our Θ's, not the Θ's themselves
- We thus face an **aggregation/"missing intercept"** problem
  - Estimand of micro experiment is a **direct** ("**partial equilibrium**") response, and so free of the general equilibrium effects that enter the full response Θ
    Why missing intercept? looking across agents absorbs common aggregate effects = enters fixed effect in cross-sectional regressions on aggregate shock. Will discuss more later.
  - Concrete example: direct response to lump-sum stimulus check **misses** tax financing, price effects (interest rates, relative goods prices), Keynesian multiplier, ...
- Will now discuss **two possible solutions**: [illustrate both for stimulus check application]
  1. Macro as explicitly aggregated micro
  2. Different shocks share identical GE effects (e.g., a common "demand multiplier")

Roadmap

- We'll begin by formalizing the **aggregation problem**
  - Using sequence-space techniques: to first order, you can generally write

    total causal effect = **micro estimate** + **GE effects**

  - We will illustrate this decomposition in a model of stimulus checks. Will be straightforward to see that such decompositions are very general, though also discuss limitations.
- We will then discuss our two **solution strategies**
  - Both will have a similar flavor: use model structure to arrive at results of the form

    total causal effect = **micro estimate** + **something that is (cleanly?) measurable**

  - The approaches will differ in how general-purpose they are/how much model structure is imposed/how easy it is to measure the **GE component**

Outline

1. The Aggregation Problem

2. Solutions
   - Macro As Explicitly Aggregated Micro
   - Commonality in GE: "Demand Equivalence"

3. Bonus: Channel Decompositions

4. Course Synthesis

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A model of stimulus checks

- The next slide will provide a brief sketch of a **rich structural GE model**
  - Model blocks: heterogeneous households ["HANK"], heterogeneous firms [Khan-Thomas, Ottonello-Winberry], fiscal & monetary policy
  - Solution method: linearization + perfect foresight [= sequence-space]
  - Policy experiment: date-0 stimulus check, financed with future taxes
- Why do I begin with a model? The **purpose of the model** is twofold:
  1. First: provide an explicit worked-out example of the decomposition

     total causal effect = micro estimate + GE effects

     Decompositions of this form are very general, as stated before. Just present one example here.
  2. Later: formally justify our two approaches to solving the aggregation problem
     The first approach is general-purpose, the second one is tailored to the application at hand.

A model of stimulus checks

1. **Households**

$$\max \quad \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t u(c_{it}, \ell_{it}) \right]$$

$$\text{s.t.} \quad c_{it} + b_{it} = w_t \ell_{it} e_{it} + \tau_t + \frac{1 + i^b_{t-1}}{1 + \pi_t} b_{it-1} + d_t$$

+ borr. constraint $b_{it} \geq \underline{b}$ & ℓ_{it} is demand-determined (= **wage-NKPC**)

2. **Production**: ≈ canonical heterogeneous-firm model, e.g., see Ottonello-Winberry (2020)

$$\max \quad \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \left(\prod_{s=0}^{t-1} \frac{1 + \pi_{s+1}}{1 + i^b_s} \right) d_{jt} \right]$$

$$\text{s.t.} \quad d_{jt} = p^l_t y(z_{jt}, k_{jt-1}, \ell_{jt}) - w_t \ell_{jt} - [k_{jt} - (1 - \delta) k_{jt-1}] - \text{adj. costs} - b_{jt} + \frac{1 + i^b_{t-1}}{1 + \pi_t} b_{jt-1}$$

+ fin. constraints on $\{b_{jt}, d_{jt}\}$ & output is demand-determined (= **price-NKPC**)

3. **Government**: spend & tax, set nominal rate (debt & monetary rules)

A PE-GE decomposition

- **Objective**: find aggregate causal effects of a transfer stimulus policy
  - Policy details: transfer path $\boldsymbol{\tau}^x = \boldsymbol{\tau}^x(\varepsilon_\tau)$ sent out to all households, financed with some future tax path $\boldsymbol{\tau}^e = \boldsymbol{\tau}^e(\varepsilon_\tau, \bullet)$, total taxes/transfers are $\boldsymbol{\tau} = \boldsymbol{\tau}^x + \boldsymbol{\tau}^e$
  - Preview: will later also look at gov't spending shock $\boldsymbol{g} = \boldsymbol{g}(\varepsilon_g)$ (+ tax financing)
- Our key tool for formalizing the aggregation problem will be a **PE-GE** decomposition of the household consumption-savings decision: [as usual: use sequence-space cons. function]

$$\widehat{\mathbf{c}}_\tau \quad \equiv \quad \mathbf{c}(\mathbf{p}_\tau, \boldsymbol{\tau}^x_\tau) - \mathbf{c}(\bar{\mathbf{p}}, \bar{\boldsymbol{\tau}}^x)$$

  - Notation: bars = steady state, hats & subscripts = IRFs, boldface = time paths
  - HH's receive **transfer** $\boldsymbol{\tau}^x$ and face **GE feedback p** ($\mathbf{i}_b$, $\boldsymbol{\pi}$, $\boldsymbol{\ell}$, $\mathbf{w}$, tax financing $\boldsymbol{\tau}^e$, ...)

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- Our key tool for formalizing the aggregation problem will be a **PE-GE** decomposition of the household consumption-savings decision: [as usual: use sequence-space cons. function]

$$\widehat{\mathbf{c}}_\tau = \underbrace{\widehat{\mathbf{c}}(\bar{\mathbf{p}}, \widehat{\boldsymbol{\tau}}^x_\tau)}_{\text{PE impact: } \widehat{c}^{PE}_\tau} + \underbrace{\widehat{\mathbf{c}}(\widehat{\mathbf{p}}_\tau, \bar{\boldsymbol{\tau}}^x)}_{\text{GE feedback: } \widehat{c}^{GE}_\tau}$$

  - Notation: bars = steady state, hats & subscripts = IRFs, boldface = paths
  - HH's receive **transfer** $\boldsymbol{\tau}^x$ and face **GE feedback p** ($\mathbf{i}_b$, $\boldsymbol{\pi}$, $\boldsymbol{\ell}$, $\mathbf{w}$, tax financing $\boldsymbol{\tau}^e$, ...)

PE impact and micro regressions

- Write out the **PE impact** as

$$\widehat{\mathbf{c}}_{\tau}^{PE} \;=\; \mathcal{C}_{\tau} \cdot \widehat{\boldsymbol{\tau}}_{\tau}^{x}$$

- $\widehat{\boldsymbol{\tau}}_{\tau}^{x}$ is a policy choice, and $\mathcal{C}_{\tau}$ is estimable from micro experiments, so we can get $\widehat{\mathbf{c}}_{\tau}^{PE}$
  - Last time we got (parts of) $\mathcal{C}_{\tau}$ from an experiment without any aggregate effects
    As far as the direct effect is concerned, (small) lottery wins = stimulus checks.
  - Straightforward exercise: can also get direct effect from an experiment with aggregate effects (like stimulus checks) if individuals differ *only* in exposure to shock
    - → In our setting: for household i corresponding to state vector (b, e) we have

$$\widehat{\mathbf{c}}_{i,\tau} \;=\; \widehat{\mathbf{c}}_{i,\tau}^{PE} \;+\; \widehat{\mathbf{c}}_{i,\tau}^{GE}$$

    - → Two households with identical state (b, e) will have identical GE effects but different PE impact if heterogenous exposure to stimulus check policy. Thus difference identifies $\widehat{\mathbf{c}}_{\tau}^{PE}$.
      This is the (idealized) interpretation of the Johnson-Parker-Souleles (2006) and Parker et al. (2013) estimand. For details see Proposition 2 and Appendix E.1 in Wolf (2023).

Aggregation

- How do we go from the estimable $\widehat{\mathbf{c}}_\tau^{PE}$ to $\widehat{\mathbf{c}}_\tau$?
- Problem: we are missing the GE term $\widehat{\mathbf{c}}_\tau^{GE}$
  - This is an aggregation problem: can use $\widehat{\mathbf{c}}_\tau^{PE}$ only if the GE term is zero (= units without a direct treatment show no overall response). Not plausible for our macro questions.
  - Also referred to as "missing intercept" problem: common GE effects are differenced out
    Stimulus check example: Keynesian demand or interest rate feedback that affects all households, independent of their direct exposure to the stimulus check policy itself.
- Next up:
  - Some quick comments on scope & limitations of this "aggregation problem" discussion
  - Two popular (classes of) solutions

A more general discussion

- This approach to **PE**-**GE** decompositions & aggregation is very general
  - Consider some general outcome of interest x. Suppose x is directly affected by a shock ε and GE "prices" p. Assume in sequence-space you can write

$$\boldsymbol{x} = \mathcal{X}(\boldsymbol{\varepsilon}, \boldsymbol{p})$$

  - Then we get the decomposition

$$\widehat{\boldsymbol{x}} = \underbrace{\mathcal{X}_{\varepsilon} \times \boldsymbol{\varepsilon}}_{\text{PE impact}} + \underbrace{\mathcal{X}_{p} \times \widehat{\boldsymbol{p}}}_{\text{GE feedback}}$$

  - Can for example easily see how investment subject to tax incentives q fits into this:

$$\widehat{\boldsymbol{i}} = \underbrace{\mathcal{I}_{q} \times \widehat{\boldsymbol{q}}}_{\text{PE impact}} + \underbrace{\mathcal{I}_{w} \times \widehat{\boldsymbol{w}} + \mathcal{I}_{r} \times \widehat{\boldsymbol{r}} + \ldots}_{\text{GE feedback}}$$

- Important **limitation**: assumes price-taking behavior

  Can still get sequence-space representations of this form with strategic interactions. But then connection to empirical measurement less clear. Example of "keeping up with the Joneses" on the board.

Other examples

Many other **famous cross-sectional studies** face the same problem:

1. Regional vs. aggregate fiscal multipliers
   Nakamura & Steinsson, Chodorow-Reich
2. China shock (regional import competition & employment)
   Autor, Dorn & Hanson
3. Bank lending cuts to firms
   Chodorow-Reich, Herreño
4. Consumption responses to stock market gains
   Chodorow-Reich, Nenov & Simsek
5. ...

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Solutions

- Standard approach: match micro estimate in **fully specified structural model**, then use that model for aggregation
  - Often not easy to see: what features of the model matter most for **PE**-**GE** mapping?
  - Attractive general-purpose diagnostic: sensitivity of GE cnfct'l to model estimation targets
    See Andrews-Gentzkow-Shapiro (2017, 2020). Not particularly common in macro (yet).
- Recently popular **semi-structural** alternative: impose enough structure so that the "missing intercept" (e.g., $\widehat{\mathbf{c}}_\tau^{GE}$) becomes somehow directly measurable
- I'll here review two examples of this, both for the **stimulus check application**:
  1. Consider a model in which the micro estimates at the same time also pin down $\widehat{\mathbf{c}}_\tau^{GE}$ ("macro as explicitly aggregated micro") Auclert, Rognlie & Straub (2024)
  2. Consider a model in which more readily available time series evidence on other shocks simultaneously pins down $\widehat{\mathbf{c}}_\tau^{GE}$ ("commonality in GE") Wolf (2023)

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Aggregation via the IKC

- Our model so far was quite rich: capital, firm-level financial frictions, general monetary rule, partially flexible prices & wages, ...
- Let's now simplify to return to the **IKC model** covered in Lecture Note 11:
  - Adapted to the notation of our big model, the equilibrium path $\widehat{\boldsymbol{c}}_\tau$ solves

$$\mathcal{C}_y \widehat{\boldsymbol{c}}_\tau + \mathcal{C}_\tau(\widehat{\boldsymbol{\tau}}^x + \widehat{\boldsymbol{\tau}}^e) = \widehat{\boldsymbol{c}}_\tau$$

  - Thus we have

$$\begin{aligned} \widehat{\boldsymbol{c}}_\tau^{PE} &= \mathcal{C}_\tau \widehat{\boldsymbol{\tau}}^x \\ \widehat{\boldsymbol{c}}_\tau^{GE} &= [I - \mathcal{C}_y]^{-1} \mathcal{C}_\tau \left(\widehat{\boldsymbol{\tau}}^x + \widehat{\boldsymbol{\tau}}^e\right) - \mathcal{C}_\tau \widehat{\boldsymbol{\tau}}^x \end{aligned}$$

- The direct effect is $\mathcal{C}_\tau \cdot \widehat{\boldsymbol{\tau}}^x$. But micro data can (in principle) get all of $\mathcal{C}_y/\mathcal{C}_\tau$ and so (through the model structure) the full GE counterfactual

  Interpretation: $\widehat{\boldsymbol{c}}_\tau$ will reflect (i) direct effect, (ii) fiscal financing rule, and (iii) Keynesian GE multiplier

Aside: getting all of $\mathcal{C}_y/\mathcal{C}_\tau$

- **Empirical evidence** so far: mostly on first column of $\mathcal{C}_y/\mathcal{C}_\tau$

  → Main takeaway: households on average *gradually* spend (small) lump-sum income receipts
  Fagereng-Holm-Natvik (2021)

- What about the rest? limited empirical evidence, but strong **theoretical predictions**
  See Wolf (2025). Shape obtains in OLG & bond-in-utility. HANK is just slightly different ...

$$\mathcal{C}_\tau \approx \underbrace{\left(1-\frac{\theta}{1+\bar{r}}\right)}_{\text{MPC}} \times \begin{pmatrix} 1 & \frac{\theta}{1+\bar{r}} & \left(\frac{\theta}{1+\bar{r}}\right)^2 & \dots \\ \theta & 1 & \frac{\theta}{1+\bar{r}} & \dots \\ \theta^2 & \theta & 1 & \dots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

Note the $\approx$. In fact MPCs along the main diagonal decline somewhat, reflecting anticipation.

→ Given (i) first column (from data) + (ii) model extrapolation for rest of $\mathcal{C}_\tau/\mathcal{C}_y$ + (iii) as'n of IKC model structure, we have enough to recover $\widehat{\boldsymbol{c}}_\tau$ for *any* stimulus check policy

Discussion

+ Reduce aggregation problem to **micro measurement exercise**
  - All macro identifying assumptions are embedded in the model. Conditional on that, we only need PE matrices, in principle estimable from micro data alone
  - Note: micro experiments both give the direct effect and the way to aggregate it

- Approach requires very **strong restrictions on the model**
  - Recall the simplifying assumptions: no capital, no firm borrowing frictions, single asset, single final good, monetary authority that fixes the real rate (and issues real bonds), ...

- Actually the micro measurement exercise is too hard, thus **still need some structure**
  - Micro data are enough to get first column of $\mathcal{C}_\tau$ and $\mathcal{C}_y$, then need extrapolation via HA consumption-savings problem for rest
    [behavioral models may extrapolate very differently, myopic model example on board ...]

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Commonality in GE

- Could we use empirical evidence on the **aggregate effects of shock/policy A** to learn about the **missing GE effects** for some micro study on **policy B**?

  Could be useful if A is somehow easier to analyze than B in the aggregate time series ...

- Less abstractly: the CBO actually uses a variant of this idea for **"demand shocks"**
  - Micro causal variation can tell us how much **stimulus checks/bonus depreciation** stimulate "demand" (consumer spending/firm investment expenditure)
  - Time series literature has identified aggregate multipliers of **gov't spending increases**

  can we just combine the two?

- Now: **formalize** this idea in the general model from before, discuss **implementation** details, assess **generality & limitations** [closely following Wolf (2023)]

Identification result

Proposition ("demand equivalence")

Return to the general model from before, and consider a policy $\boldsymbol{\varepsilon_\tau}$. *Suppose that:*

A1 ...

A2 ...

A3 ...

If a fiscal spending policy shock $\boldsymbol{\varepsilon_g}$ *is s.t. (i)* $\widehat{\boldsymbol{g}}_g = \widehat{\boldsymbol{c}}_\tau^{PE}$ *and (ii)* $\widehat{\boldsymbol{\tau}}_g^e = \widehat{\boldsymbol{\tau}}_\tau^e$, *then, to first order,*

$$\widehat{\boldsymbol{c}}_\tau = \underbrace{\widehat{\boldsymbol{c}}_\tau^{PE}}_{\textbf{PE response}} + \underbrace{\widehat{c}_g}_{\textbf{GE feedback}}$$

Identification result

Proposition ("demand equivalence")

Return to the general model from before, and consider a policy $\boldsymbol{\varepsilon_\tau}$*. Suppose that:*

A1 *Households and gov't consume a* **common final good**.

This assumption has no bite in our model since it only features one good anyway. I'm stating it just to make clear the importance of this restriction.

A2 *Households and gov't borrow and lend at* **identical rates**.

This assumption also has no bite in our model (one asset, no borr. wedge). Again stating for emphasis.

A3 *There are* **no wealth effects** *in labor supply and/or wages are fully* **rigid**.

If a fiscal spending policy shock $\boldsymbol{\varepsilon_g}$ *is s.t. (i)* $\widehat{\boldsymbol{g}}_g = \widehat{\boldsymbol{c}}_\tau^{PE}$ *and (ii)* $\widehat{\boldsymbol{\tau}}_g^e = \widehat{\boldsymbol{\tau}}_\tau^e$*, then, to first order,*

$$\widehat{\boldsymbol{c}}_\tau \quad = \quad \underbrace{\widehat{\boldsymbol{c}}_\tau^{PE}}_{\textbf{PE response}} \quad + \quad \underbrace{\widehat{\boldsymbol{c}}_g}_{\textbf{GE feedback}}$$

Proof sketch

- Equilibrium = solution to many mkt-clearing conditions + other restrictions
  [output, gov't budget, asset markets, labor, equity valuation, asset arbitrage, Taylor rule, ...]

$$\underbrace{\mathbf{H}(\mathbf{p};\boldsymbol{\varepsilon})}_{\text{full equilibrium system}} \quad = \quad \mathbf{0}$$

- Proof strategy: identical **excess demand/supply** *in all markets*

  A1 **Output**: by property (i), identical demand pressure for common final good

$$\mathbf{c}(\mathbf{p};\boldsymbol{\varepsilon}) + \mathbf{g}(\boldsymbol{\varepsilon}) \;=\; \mathbf{y}(\mathbf{p};\boldsymbol{\varepsilon}) - \mathbf{i}(\mathbf{p};\boldsymbol{\varepsilon})$$

  A2 **Gov't budget**: by property (ii), identical tax financing for transfers & gov't spending

$$\boldsymbol{\tau}^e \;=\; \boldsymbol{\tau}^e(\mathbf{p};\boldsymbol{\varepsilon})$$

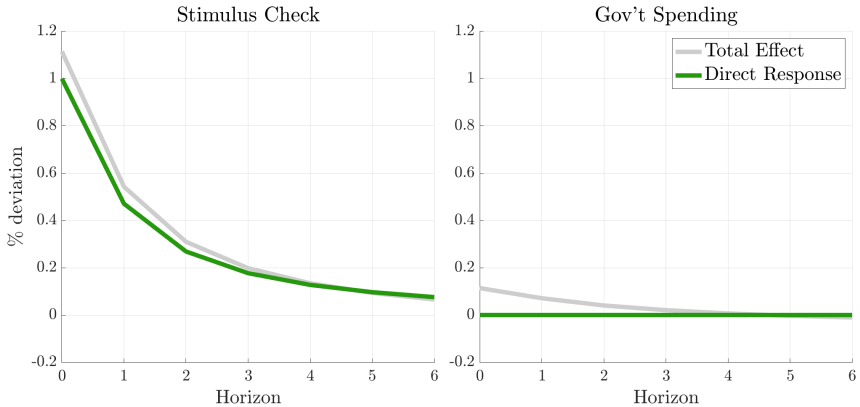
  Note: without A2 we couldn't ensure property (ii).

  A3 **Labor**: no shift in labor supply/shift is irrelevant

$$\boldsymbol{\ell}^h(\mathbf{p};\boldsymbol{\varepsilon}) \;=\; \boldsymbol{\ell}^f(\mathbf{p};\boldsymbol{\varepsilon})$$

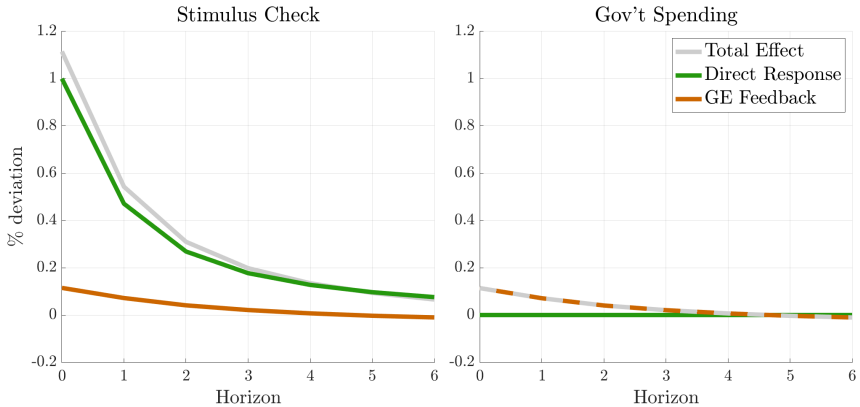
Numerical illustration

Let's now see this result in action in a sticky-wage HANK model ...



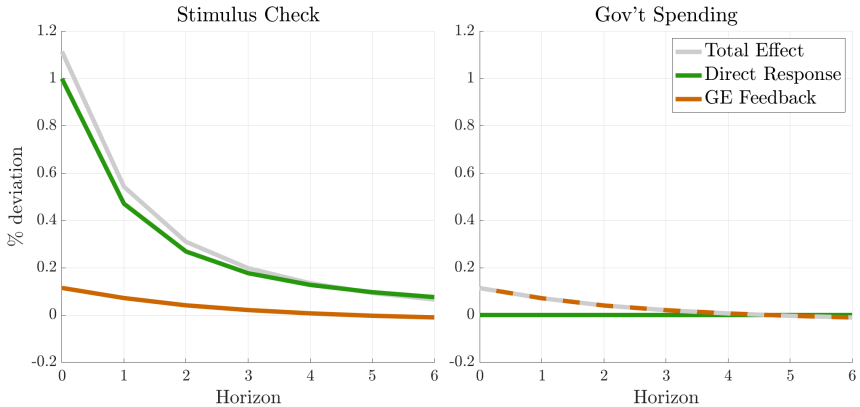
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Let's now see this result in action in a sticky-wage HANK model ...



Numerical illustration

Let's now see this result in action in a sticky-wage HANK model ...



We can leverage this result to operationalize the CBO intuition ...

Measurement inputs

Result suggests that we can combine **cross-sectional** and **time-series** measurement:

1. **Cross-sectional** identification

$$\boldsymbol{\beta}_\tau \times \varepsilon_{\tau t} \quad = \quad \widehat{\mathbf{c}}_\tau^{PE}$$

   Get $\boldsymbol{\beta}_\tau \times \varepsilon_{\tau t}$ either from pure micro regression (like lottery wins) or leveraging heterogeneous direct exposure to stimulus check experiment, à la Johnson-Parker-Souleles (2006) or Parker et al. (2013).

2. Fiscal **time series** experiments
   - We have seen several time series approaches to identifying fiscal spending shocks and so estimating $\widehat{c}_g$. E.g.: (i) narrative, (ii) forecast errors, (iii) timing/exclusion restrictions Blanchard-Perotti (2002), Mountford-Uhlig (2009), Ramey (2011), Caldara-Kamps (2017), ...
   - Summary: most estimates lie "in a fairly narrow range, 0.6 to 1" Ramey (2019)

$$\underbrace{\qquad\qquad}_{\text{combine them to arrive at } \widehat{c}_\tau}$$

Matching experiments

Challenge: necessary condition to combine experiments is identical net excess demand — $\widehat{\boldsymbol{g}}_g = \widehat{\boldsymbol{c}}_\tau^{PE}$. Given $\widehat{\boldsymbol{c}}_\tau^{PE}$, how can we find the required gov't spending shock?

- Natural approach: look for best **linear combination**
  - Suppose time series analysis has identified several shocks with paths $\widehat{\mathbf{g}}_{g_k}$. Projection:

$$\widehat{\mathbf{c}}_\tau^{PE} \quad = \quad \sum_{k=1}^{n_k} \gamma_k \times \widehat{\mathbf{g}}_{g_k} \quad + \quad \text{error}$$

  - Then

$$\sum_{k=1}^{n_k} \gamma_k \times \widehat{\mathbf{c}}_{g_k}$$

    promises to capture general equilibrium effects *up to the error term*
  - Note that this is the same multi-shock idea as what we used for policy rule counterfactuals!

Implementation & interpretation

- Suppose you have ensured that $\widehat{\boldsymbol{g}}_g = \widehat{\boldsymbol{c}}_\tau^{PE}$. Now construct

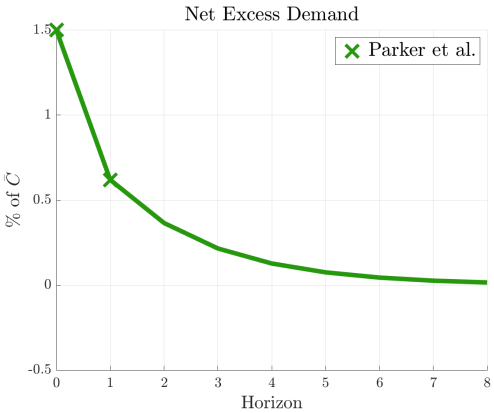
$$\widehat{\mathbf{c}}_\tau \quad = \quad \underbrace{\widehat{\mathbf{c}}_\tau^{PE}}_{\text{PE response}} \quad + \quad \underbrace{\sum_{k=1}^{n_k} \gamma_k \times \widehat{\mathbf{c}}_{g_k}}_{\text{GE feedback}} \tag{1}$$

How should we interpret (1)? Valid **GE** counterfactual for stimulus check shock ε_τ s.t.:

a) ε_τ and ε_g are associated with the same movements in taxes (recall: condition (ii) in th'm)

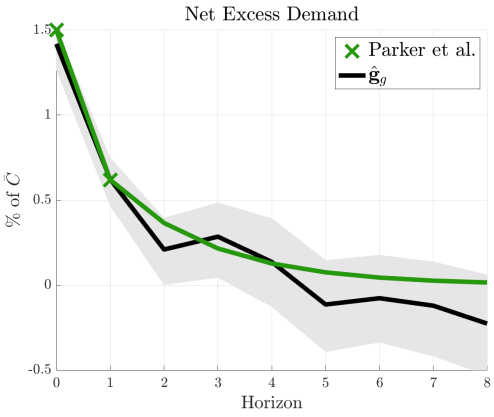
b) ε_τ and ε_g occur in the same macro environment (same policy regime, cyclical state, ...)
Why? recall that th'm used (log-)linearization

Application: PE demand



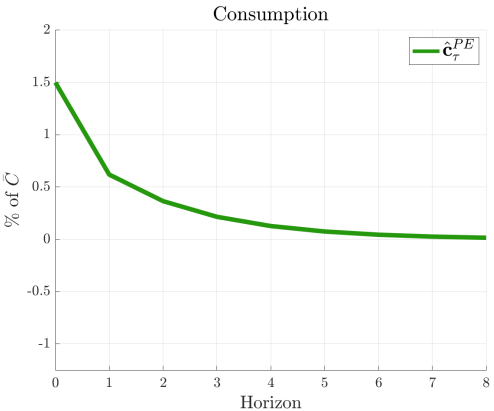
- Cross-sectional identification
  [Parker-Souleles-Johnson-McClelland (2013)]
  - Experiment: one-off stimulus checks (around $600 per household)
  - Find: strong, very short-lived response

Application: PE demand



- Cross-sectional identification
  [Parker-Souleles-Johnson-McClelland (2013)]
  - Experiment: one-off stimulus checks (around $600 per household)
  - Find: strong, very short-lived response
- Time series identification [Ramey (2011)]
  - Identification as'n: forecast errors as IV
  - Find: short-lived uptick in g, deficit-financed, muted interest rate response

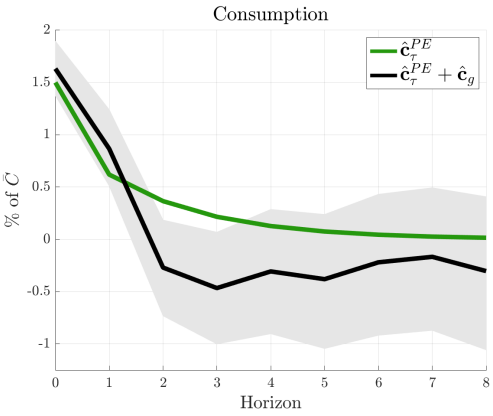
Application: GE counterfactual



- Aggregate via demand equivalence:

$$\widehat{\mathbf{c}}_\tau = \underbrace{\widehat{\mathbf{c}}_\tau^{PE}}_{\text{PE response}} + \underbrace{\widehat{\mathbf{c}}_g}_{\text{GE feedback}}$$

Application: GE counterfactual



- Aggregate via demand equivalence:

$$\widehat{\mathbf{c}}_\tau = \underbrace{\widehat{\mathbf{c}}_\tau^{PE}}_{\text{PE response}} + \underbrace{\widehat{\mathbf{c}}_g}_{\text{GE feedback}}$$

- Main result: strong impact stimulus with

full C response $\approx$ **micro estimate**

- Interpretation: same conclusion in *any* model s.t.:
  - (i) demand equivalence holds
  - (ii) micro & macro moments are matched

Assessment

How should we interpret these results?

- So far: valid counterfactuals for *any model* that satisfies **demand equivalence**
  demand equivalence + cross-sectional & time series moments from literature ⇒ will necessarily arrive at our results (= "sufficient statistics" in public finance)
- But the required assumptions are of course quite **strong**
- **Q**: what does it look like in models that break exact demand equivalence?
  - **A**: probably miss some GE crowding-out = upward-biased Details

Discussion

+ Applies for a large **space of models**
  - Allows for investment (+ very general firm block) & general monetary rule, though still requires single good & no borrowing frictions
+ Implementable purely through **empirical measurement**
- **Not always applicable**
  - Need to find experiments so that net excess demand paths are aligned
  - Works only for demand shocks, so not a general-purpose aggregation methodology
    More generally: match shocks as wedges (here "excess demand"). Ideas for other applications?
- Need to **trust time series evidence** for a different shock
  - Requires relatively greater confidence in time series estimates for fiscal spending experiment than the demand shock of interest

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Micro data & transmission channels

- Identified aggregate shocks ε_t give us aggregate IRFs Θ. Natural **Q**: what are the "**channels**" through which ε_t transmits to y_t?
- Could formalize the concept of channels through aggregate decision rules, as we did in the "aggregation problem" discussion. Returning to our running consumption example:

$$\boldsymbol{c} = \mathcal{C}(\boldsymbol{r}, \boldsymbol{y}) \quad \Rightarrow \quad \widehat{\boldsymbol{c}} = \mathcal{C}_r \widehat{\boldsymbol{r}} + \mathcal{C}_y \widehat{\boldsymbol{y}}$$

  Then ε transmits to c via two channels:

$$\widehat{\boldsymbol{c}}_\varepsilon = \underbrace{\mathcal{C}_r \widehat{\boldsymbol{r}}_\varepsilon}_{\text{interest rate channel}} + \underbrace{\mathcal{C}_y \widehat{\boldsymbol{y}}_\varepsilon}_{\text{income channel}}$$

- Immediate that micro + macro data *together* can identify the **split into channels**: the time series gives $\widehat{\boldsymbol{c}}_\varepsilon$, $\widehat{\boldsymbol{r}}_\varepsilon$ and $\widehat{\boldsymbol{y}}_\varepsilon$, while the cross section gives $\mathcal{C}_y$ and $\mathcal{C}_r$

Application: direct vs. indirect effects of monetary policy

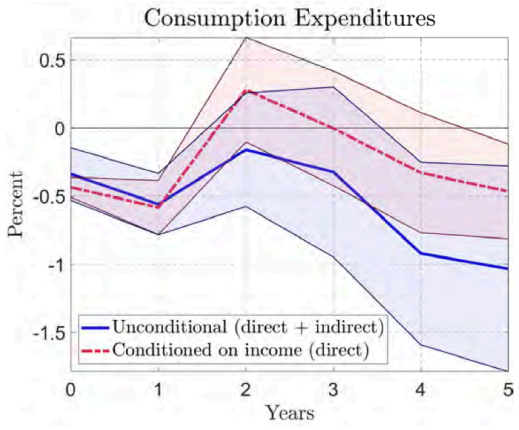
- Recall **main message** of Kaplan-Moll-Violante (2018):
  - Monetary policy transmits to consumption via direct effects [interest rates] and indirect general equilibrium effects [chiefly income movements]
  - In HANK models: mostly indirect effects, while the opposite is true in RANK models
- **Empirical analysis** [Holm-Paul-Tischbirek (2021)]
  - Get $\widehat{\boldsymbol{c}}_\varepsilon$ and $\widehat{\boldsymbol{y}}_\varepsilon$ from time-series analysis. Then use lottery win evidence for $\mathcal{C}_y$. From here can decompose full consumption impulse response into income channel and remainder.
  - But we only have part of $\mathcal{C}_y$ – what to do? Can show: their econometric strategy is equivalent to assuming *myopia*, i.e., setting

$$\mathcal{C}_y(\bullet, h) = \begin{pmatrix} \mathbf{0}_{h-1} \\ \mathcal{C}_y(\bullet, 1) \end{pmatrix}$$

    where $\mathcal{C}_y(\bullet, 1)$ comes from the empirical lottery evidence

    Can interpret this as the "channel" analogue of Sims & Zha! Everything indeed hangs together.

Application: direct vs. indirect effects of monetary policy



Why do we care about channels?

- Aside: why should we even care about this **decomposition** into **channels**?
- In our linear SVMA framework:
  - Key objective of policy analysis is policy cnfct'l evaluation. We saw that, for this purpose, the full matrix of policy causal effects Θ_ν – and not their decomposition – suffices.
  - So the question is: does matching "channels" change extrapolation of policy shock IRFs across horizons? For HANK vs. RANK the answer is "no," as we've seen.
- Going **beyond linearity**:
  - Empirical evidence only gives us policy shock causal effects for certain business-cycle conditions, of certain signs, and of certain sizes [remember Lecture Note 7]
  - Maybe getting the channels right matters for extrapolation not across *time* [i.e., completing Θ_ν, see above], but for *state*, *sign*, or *size*? good open **Q**!

Outline

1. The Aggregation Problem

2. Solutions
   - Macro As Explicitly Aggregated Micro
   - Commonality in GE: "Demand Equivalence"

3. Bonus: Channel Decompositions

4. Course Synthesis

Course synthesis: identification in macroeconomics

$$y_t = \sum_{\ell=0}^{\infty} \Theta_\ell \varepsilon_{t-\ell}$$

- Two sources of empirical **identifying information**:
  1. **Aggregate time series**: identified shocks/policy treatments via invertibility + restrictions or macro IVs, estimated using LPs/VARs
  2. **Cross-sectional micro data**: slopes of behavioral blocks ($\mathcal{C}_\tau$, $\mathcal{I}_r$, ...), several steps removed from the Θ's
- Two key roles for **theory**:
  1. **Identification results** use the empirical evidence as *sufficient statistics*: systematic policy rule counterfactuals from policy shocks, GE counterfactuals from PE slopes
  2. **Extrapolation & aggregation** where the evidence ends – residual role of model structure

Thank you!

Let's please keep in touch as you progress with your research careers. I had a great time teaching you and hope you picked up something useful from this class ☺

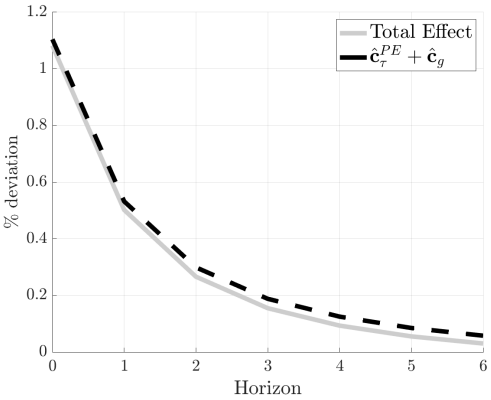
Appendix

Demand equivalence: accuracy

- Idea: in models that **break equivalence** compute the plim of aggregation procedure
  1. Estimated HANK model [Smets-Wouters (2007) + Kaplan-Moll-Violante (2018)]
     Note: breaks equivalence *only* via labor assumption
  2. Further extensions that jointly break asn's 1-3
- **Main result**: miss some **GE crowding-out**

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Estimated HANK Model



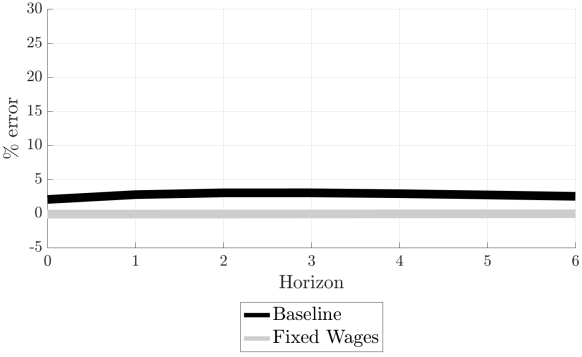
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- Estimate HANK model, solve at posterior mode
  Results similar across the posterior distribution
- First finding: nearly exact in rich estimated model
- What's going on?
  - Equivalence fails *only* due to labor channel

$$\widehat{\mathbf{c}}_\tau = \widehat{\mathbf{c}}_\tau^{PE} + \widehat{\mathbf{c}}_g + \text{error}\left(\widehat{\boldsymbol{\ell}}_\tau^{PE}\right)$$

  - But wages are quite sticky ($\phi_w = 0.6$)
  - Well-known: for transitory fluctuations, with sticky prices/wages, labor wedge shocks matter little
    Christiano (2011, 2012), Auclert et al. (2020)

Further extensions



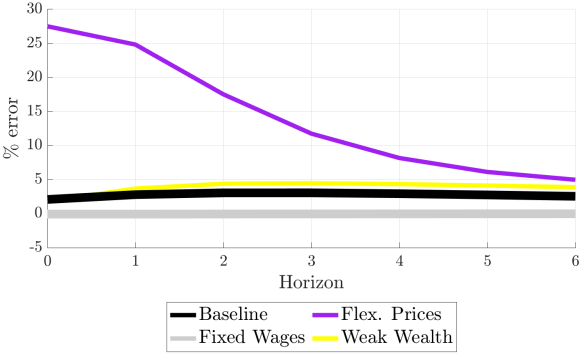
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- For several further model variants compute

$$\text{error} \quad = \quad \frac{(c_\tau^{PE} + c_g) - c_\tau}{c_{\tau,0}}$$

- Check violations of each key assumption:

Further extensions



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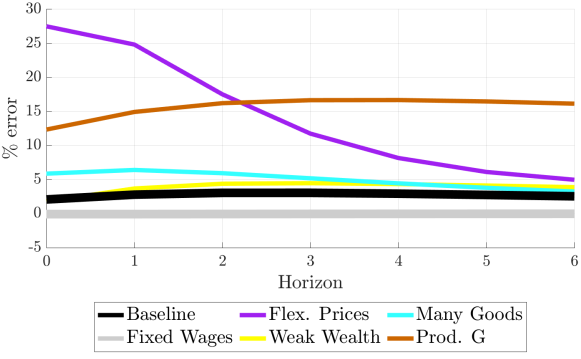
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back

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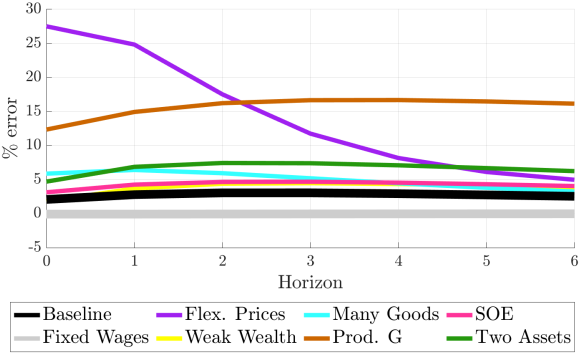
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- Check violations of each key assumption:

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  A2 multiple goods, gov't investment

Further extensions



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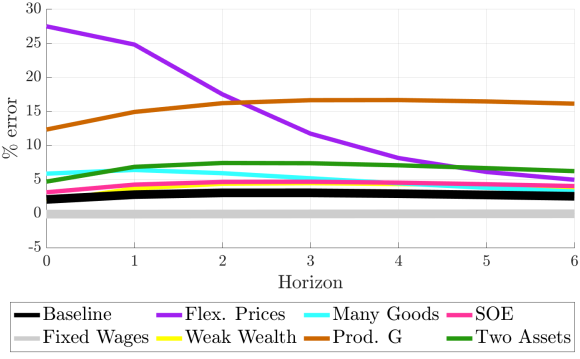
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  A2 multiple goods, gov't investment

  A3 multiple assets, small open economy

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Further extensions



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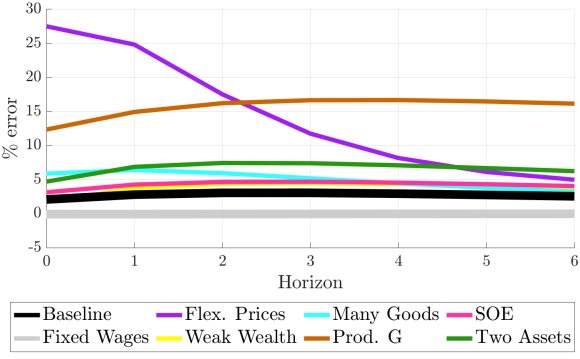
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main message: approximation is biased *up*

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back

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- Check violations of each key assumption:

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  main message: approximation is biased *up*

- But: bound likely to be tight for checks

  small & transitory, little evidence of labor adjustment, do not use gov't investment, ...